ModelIt!

Teacher's Guide

Can You Bend Water?

Exploring the Properties of Matter

Quick Reference

Grade Level	Grade 2
NGSS Standard	2-PS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 2-PS1-1

Plan and conduct an investigation to describe and classify different kinds of materials by their observable properties.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students investigate how water changes shape by pouring it into different containers and recording their observations.
Disciplinary Core Idea	**PS1.A Structure and Properties of Matter** — Different kinds of matter exist and many of them can be either solid or liquid depending on temperature. Matter can be described and classified by its observable properties.
Crosscutting Concept	**Patterns** — Students observe the pattern that liquids always take the shape of their container while solids do not.

CAST Testing Alignment

MULTIPLE CHOICE:

You pour water from a tall skinny glass into a wide flat bowl. What happens to the shape of the water?

CONSTRUCTED RESPONSE:

Explain how you can tell the difference between a solid and a liquid by observing their properties. Use an example from your investigation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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> **For Your Confidence**
>
> This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Matter is everything around us — the air we breathe, the water we drink, and the ground we walk on. Understanding the properties of matter helps young scientists classify and describe the world.

What Is Matter?

Matter is anything that takes up space and has weight. Everything you can see, touch, and feel is made of matter — your desk, your pencil, the air around you, and even you! Scientists sort matter into groups based on properties we can observe, like shape, color, texture, and whether it flows.

Solids and Liquids

Solids have a fixed shape. A crayon stays crayon-shaped whether it is on your desk or in your backpack. Liquids flow and take the shape of whatever container they are in. Water in a cup is cup-shaped, but the same water poured into a bowl becomes bowl-shaped. The amount of matter does not change — only the shape does.

Why Do Liquids Flow?

Inside every liquid, tiny pieces of matter (too small to see) are constantly moving and sliding past each other. Because these tiny pieces are not locked in place, the liquid can flow and fill any shape. In a solid, the tiny pieces are locked tightly together, so the solid keeps its shape.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify container shape as an external component and water shape and water level as internal components that change in response to the container.

E **Establish Relationships**
Students determine that changing the container shape causes the water shape to change (positive relationship), and that a wider container means a lower water level for the same amount of water.

V **Visualize & Model**
Students build a model showing how pouring water into different containers changes its shape but not the total amount.

E **Evaluate Outcomes**
Students run tall cup, wide bowl, and round jar scenarios to see how the same water looks different in each container.

R **Revise & Extend**
Students add temperature or amount of water to explore how other factors affect what they observe.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Who here has ever tried to grab water? What happened? Could you hold it in your hands?"*

Watch For

Let students share. If possible, have a small tub of water for a quick demo.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to figure out something really cool about water — can it change shape?"*

Watch For

Read objectives aloud together. Introduce the words matter, property, liquid, and solid with hand motions.

Slide 3: The Big Question

3 min

Say: *"Here is our big question: Can you bend water? What do you think?"*

Watch For

Thumbs up or thumbs down: Can you bend water? Have 2-3 students share their thinking.

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build a model to figure this out — step by step, like real scientists."*

Watch For

Preview each step briefly. Remind students that models help us understand how things work.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What are the parts of our experiment? Which part do WE control, and which parts change on their own?"*

Watch For

Guide students to sort Container Shape as external (we pick it) and Water Shape and Water Level as internal (they change by themselves).

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When we change the container shape, what happens to the water shape? Does it change or stay the same?"*

Watch For

Help students draw arrows from Container Shape to Water Shape and Water Level. Discuss whether the change is positive or negative.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Time to test! Let us start by pouring water into a tall skinny cup. What do you think will happen?"*

Watch For

Have students predict first, then run each scenario. Use real containers if possible. Compare tall cup vs. flat bowl vs. round jar.

Slide 8: What Did We Discover?

7 min

Say: *"So can you bend water? Not exactly — but water does something even cooler. It FLOWS into any shape!"*

Watch For

Review discoveries together. Compare water to a solid object like a block. Have students share one thing they learned with a partner.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Container Shape
- **INTERNAL:** Water Shape, Water Level

Relationships

- **Container Shape to Water Shape:** POSITIVE (+) — When you change the container shape, the water shape changes to match it. A tall container makes the water tall; a flat container makes the water flat. The water always copies the container.
- **Container Shape to Water Level:** NEGATIVE (-) — When the container gets wider, the water level gets lower because the same amount of water spreads out more. A narrow container pushes the water up higher.

Key Discoveries

- Water always takes the shape of its container — it does not have its own fixed shape
- The amount of water stays the same even when the shape changes
- Solids keep their shape, but liquids flow and fill the space they are given
- Properties like shape help us tell the difference between solids and liquids

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Water disappears when you pour it out"	Water does not disappear — it moves! When you pour water from one container to another, all the water is still there. It just changed shape and location. If water spills, it spreads out on the surface but it is still there.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Water has its own shape"	Water does NOT have a fixed shape like a block or a ball. Water is a liquid, which means it takes the shape of whatever container holds it. If there is no container, water flows and spreads out flat because of gravity.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bigger containers have more water"	A bigger container does not mean more water. The same amount of water just looks different in different containers. In a tall skinny cup the water looks like a lot because it is high, but in a wide bowl it looks like less even though it is the same amount.